

Variance Retention: A Forecast-Verification Diagnostic for Skilful-but-Smoothed Point Forecasts of Daily PM₁₀

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Abstract

Verification of deterministic air-quality forecasts relies heavily on persistence-relative RMSE skill, yet a forecast can be certified as skilful while its trajectory is dynamically over-smoothed. We introduce variance retention as a complementary verification diagnostic and quantify this skill-fidelity tension in a rolling-origin multi-horizon evaluation ($h = 1, \dots, 7$) over 17 heterogeneous stations from the Spanish MITECO monitoring network, using persistence as an explicit baseline.

Five model families are compared: two direct lag-only machine-learning forecasters (HGB direct and Ridge direct), a classical statistical baseline (SARIMA), a variance-preserving reference (seasonal naive), and a decomposition-based control (STL+Ridge). Direct HGB and Ridge achieve positive RMSE skill in all station-horizon cells and are frequently significantly better than persistence under BH-adjusted Diebold–Mariano tests, yet they exhibit near-universal variance collapse ($\alpha < 0.5$) in 118/119 cells each. SARIMA partially reproduces the same pattern, collapsing in 110/119 cells. Seasonal naive preserves variance but does not improve RMSE on median, whereas STL+Ridge preserves or inflates variance and attains high exceedance recall at the cost of poor RMSE skill and many false alarms.

The results show that maximising RMSE skill alone is insufficient for operational use: performance must be diagnosed jointly in terms of baseline-relative accuracy, dynamic fidelity (variance retention), and event-detection behaviour.

Keywords: forecast verification, deterministic point forecasts, PM₁₀ forecasting, rolling-origin evaluation, persistence baseline, variance retention, exceedance diagnostics

1. Introduction

Daily PM₁₀ forecasts support a short-horizon interpretation in air-quality management and public-health communication, where decisions are sensitive to near-term changes as well as average error [1, 2]. In operational settings, however, usefulness depends not only on average error but also on whether forecasts reproduce meaningful day-to-day changes and identify episodes.

In multi-horizon settings, persistence-relative RMSE skill is a necessary summary because it tests whether a model improves over an explicit operational baseline [3]. However, point-forecast accuracy summaries remain incomplete for decision support [4]: a model can be significantly skilful in RMSE while producing trajectories that are dynamically over-smoothed.

To address this gap, we introduce variance retention as a post-evaluation diagnostic layer to read dynamic fidelity alongside RMSE-based skill in a rolling-origin daily PM₁₀ study. We complement the joint skill-fidelity reading with (i) bootstrap intervals for the variance-retention ratio α , (ii) threshold-sensitivity

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checks for the collapse convention, (iii) BH-adjusted Diebold–Mariano tests against persistence, (iv) exceedance diagnostics for episode sensitivity and false alarms, and (v) a Murphy decomposition to connect the empirical patterns to forecast verification theory [5].

We further ask whether the skill–fidelity tension is specific to direct lag-only ML forecasters or also appears in classical statistical and decomposition-based alternatives.

Contributions. This paper contributes to forecast-verification practice, not to forecasting methodology or atmospheric-process understanding. Specifically: (i) we formalise *variance retention* $\alpha(h)$ as a bounded, post-hoc verification diagnostic that complements persistence-relative skill by exposing amplitude collapse in deterministic point forecasts, together with an interpretive skill–retention template (Table 1); (ii) across 17 heterogeneous PM₁₀ stations and five model families, we show that statistically significant RMSE gains over persistence (BH-adjusted Diebold–Mariano) routinely coexist with near-universal variance collapse and degraded event detection, and we connect this pattern to a Murphy decomposition of mean squared error; and (iii) we distil a minimum set of reporting practices for verifying deterministic air-quality forecasts. The claims are deliberately bounded to the evaluated setting and are not offered as a general theory of forecast skill.

2. Evaluation Reporting Gaps in Particulate Matter Forecasting

Machine-learning and deep-learning air-quality forecasting has grown rapidly, especially for particulate matter. Reviews describe increasingly complex architectures, hybrid modelling strategies, and data-driven pipelines for PM₁₀ and PM_{2.5} prediction [6, 7, 8].

This growth has shifted much of the literature toward comparisons between modelling architectures, feature sets, and learning algorithms. However, methodological claims about “predictive gains” are only interpretable when the evaluation design is stated clearly enough that a reader can understand what was tested, what the baseline comparator was, and how the forecast horizons were handled [9, 10].

Five evaluation reporting dimensions are central to operational interpretability in particulate matter forecasting: temporal validation, leakage-safe preprocessing, persistence or naive baselines, horizon-wise reporting, and operational-usefulness framing. Temporal validation establishes deployment-like chronological credibility; leakage-safe preprocessing protects against information bleed across train–test boundaries; persistence baselines provide a minimum operational comparator; horizon-wise reporting supports multi-step interpretability; and operational framing links performance summaries to decision relevance [11, 12, 13].

To motivate why an additional trajectory-fidelity diagnostic can be useful in practice, we conducted a PRISMA 2020 audit of PM₁₀/PM_{2.5} forecasting studies (Scopus and Web of Science, 2015–2025; reproducible materials archived at Zenodo, concept DOI: 10.5281/zenodo.18675211, release: 10.5281/zenodo.19125915). At the abstract level, key evaluation choices (e.g., explicit persistence baselines, horizon-wise reporting, and temporally appropriate validation) were often not made visible. These rates reflect reporting opacity rather than confirmed absence. The need for such a diagnostic is amplified by a broader reporting problem in PM forecasting evaluation: when evaluation details and trajectory behaviour are not reported, baseline-relative improvements can be difficult to interpret operationally.

3. Post-Evaluation Diagnostic Framework

The post-evaluation framework is organised around three complementary questions:

Skill: Does the model beat persistence? In this manuscript, skill is defined relative to persistence under RMSE, so positive $\text{Skill}(h)$ indicates lower RMSE than persistence at the same horizon h [3, 11].

Variance retention: Does the forecast retain observed variability? The variance-retention ratio $\alpha(h) = \text{Var}(\hat{y})/\text{Var}(y)$ is interpreted post-evaluation; low $\alpha(h)$ indicates variability shrinkage consistent with smoothed trajectories, which can complicate operational interpretation even when RMSE skill is positive.

Skill_{VP}: What compact auxiliary summary helps read both together? Skill_{VP} is reported only as a bounded diagnostic auxiliary alongside Skill(h) and $\alpha(h)$, not as a replacement for standard accuracy metrics.

Table 1: Interpretive template for joint reading of persistence-relative skill and variance retention. Rows distinguish positive from negative skill; columns distinguish retained from collapsed variance. The template is a diagnostic aid, not a decision rule.

	$\alpha(h) \geq 0.8$ (retained)	$\alpha(h) < 0.5$ (collapsed)
Skill(h) > 0	Desirable case. Model improves over persistence and preserves observed variability. Forecasts are more credible under both diagnostics.	Skill-fidelity decoupling. Model reduces RMSE but dampens day-to-day fluctuations. Forecasts track the mean level but may under-represent episode amplitude.
Skill(h) $\leq$ 0	No improvement with preserved dynamics. Model does not outperform persistence despite retaining variance. Forecasts reproduce variability but are not more accurate.	Worst case. Model is both less accurate than persistence and smoothed. No evident operational value under these two diagnostics.

The boundaries $\alpha = 0.8$ and $\alpha = 0.5$ are diagnostic conventions adopted in this study; they are not derived from a formal loss criterion.

4. Case Study and Experimental Setting

The case study uses daily PM₁₀ observations from 17 monitoring stations in the Spanish MITECO national air-quality network. The station set is intentionally heterogeneous in site type (industrial, traffic, urban/background, residential, and rural/remote), spans multiple autonomous communities (including the Comunitat Valenciana, Catalunya, Aragón, and Castilla-La Mancha), and covers the 2017–2024 period, with observational coverage ranging from 80.4% to 98.5%.

Three stations are described in detail as representative examples of the three environmental types.

- **Elx-Agroalimentari** (ES0446A): suburban/industrial station; 44 m a.s.l.; 2,350 observed daily values (80.4% coverage, 2017–2024).
- **Valencia-Vivers** (ES1619A): urban/residential station; 11 m a.s.l.; 2,679 observed daily values (91.7% coverage, 2017–2024).
- **Zarra-EMEP** (ES0012R): rural remote EMEP station; 855 m a.s.l.; 2,804 observed daily values (96.0% coverage, 2017–2024).

Meteorological covariates (temperature, relative humidity, wind speed, and wind direction) are available for Elx-Agroalimentari only but are deliberately excluded from all experiments: the study intentionally employs a lags-only design to isolate the autoregressive signal.

Each series was reindexed to a complete daily calendar before lag construction. Dates with missing PM₁₀ values, missing lagged predictors, or unavailable verification targets after horizon alignment were excluded by listwise deletion. No interpolation or imputation was applied.

The objective is not to maximise predictive accuracy, but to create a transparent evaluation setting in which persistence-relative skill, horizon-wise behaviour, and variance retention can be inspected jointly.

Five diagnostic comparators are evaluated. HGB direct and Ridge direct are direct lag-feature ML forecasters. SARIMA is included as a sparse classical statistical baseline. STL+Ridge is a decomposition-based variant used to test whether explicit seasonal separation can preserve trajectory variability. Seasonal naive is included as a variance-preserving seasonal reference. These models are not used as a leaderboard

of optimised forecasting systems, but as diagnostic contrasts for separating baseline-relative accuracy from trajectory fidelity.

Forecasts are evaluated for horizons $h = 1, \dots, 7$ using rolling-origin (time-series) evaluation with a mandatory persistence baseline, and the analysis focuses on point forecasts. When preprocessing is applicable (e.g., scaling), transformations are fitted on the training window only and then applied to the corresponding test window to reduce leakage risk [14, 12].

A lags-only design is intentionally restrictive: if positive skill appears even in this controlled setting, the variance-retention diagnostic can reveal whether the improvement is accompanied by retained forecast variability or by systematically smoothed trajectories.

5. Metrics and Diagnostic Categories

5.1. Persistence-relative skill

Let y_h denote the verifying observation at horizon h , and $\hat{y}_h$ the corresponding point forecast produced by a model. Let $\hat{y}_h^{(p)}$ denote the persistence baseline forecast for the same horizon. For a chosen loss function $L(\cdot, \cdot)$, the baseline-relative (persistence-relative) skill score at horizon h is

$$\text{Skill}(h) = 1 - \frac{\mathbb{E}[L(y_h, \hat{y}_h)]}{\mathbb{E}[L(y_h, \hat{y}_h^{(p)})]}. \quad (1)$$

In this work, the empirical results use an RMSE-based loss, so $\text{Skill}(h) > 0$ indicates a lower RMSE than persistence for the same horizon and evaluation sample [3].

Persistence-relative skill is useful because it anchors interpretation to an operationally meaningful comparator rather than to a model-to-model ranking alone. It also supports horizon-wise reading: in multi-step settings, skill can vary substantially across lead times.

At the same time, skill is a summary of pointwise error behaviour under a chosen loss. It does not, by itself, describe the trajectory shape of the point forecasts, and it does not indicate whether an apparent improvement is associated with reduced dynamic variability.

5.2. Variance retention

We define the variance-retention ratio at horizon h as

$$\alpha(h) = \frac{\sigma_{\hat{y}_h}^2}{\sigma_{y_h}^2} = \frac{\text{Var}(\hat{y}_h)}{\text{Var}(y_h)}, \quad (2)$$

the ratio of the predicted to the observed *variance* at horizon h . Both variances are population variances (ddof = 0) computed over the matched finite forecast–observation pairs within each station–model–horizon cell, so $\alpha(h)$ is the fraction of observed variance retained by the forecast (a value of 0.5 corresponds to half the observed variance retained). This is a variance ratio, not a standard-deviation ratio; the two differ by a square and are not interchangeable.

The ratio $\alpha(h)$ compares forecast trajectory variability with observed variability at the same horizon. Values $\alpha(h) \ll 1$ indicate variance shrinkage consistent with smoothed point-forecast trajectories, while values $\alpha(h) \gg 1$ indicate forecast variance inflation.

Variance retention should not be interpreted as a complete measure of trajectory quality. A forecast may retain the observed variance and still be biased, phase-shifted, or poor at reproducing individual episodes. The diagnostic is therefore intended to detect one specific failure mode—variance shrinkage—rather than to certify full dynamic adequacy.

Variance retention is not a calibration measure and is not intended to assess probabilistic uncertainty. It is a bounded post-evaluation diagnostic for reading point forecasts when operational interpretation depends on day-to-day fluctuations as well as on average accuracy [13].

5.3. Auxiliary Skill_{VP}

To summarise the joint behaviour of baseline-relative accuracy and variance retention, we also report an auxiliary variance-penalised skill diagnostic

$$\text{Skill}_{VP}(h) = \text{Skill}(h) \cdot \alpha(h). \quad (3)$$

The auxiliary diagnostic is intended to highlight cases where positive persistence-relative skill co-occurs with strong variance shrinkage. The product form is not claimed to be uniquely optimal; it is used here only as a transparent descriptive penalty that keeps the sign of Skill(h) while reducing positive skill when forecast variance is strongly collapsed. It should not be interpreted as a replacement for standard accuracy metrics, uncertainty quantification, or statistical significance testing [15].

Accordingly, Skill_{VP}(h) is used only as a bounded diagnostic layer for post-evaluation interpretation. Its role is to provide a compact joint summary alongside Skill(h) and $\alpha(h)$, rather than to define a universal performance criterion.

5.4. Diagnostic flags

In addition to $\alpha(h)$ and Skill_{VP}(h), the post-evaluation reports diagnose flags for horizon-wise interpretation. The **collapse_flag** is used when $\alpha(h) < 0.5$. The **inflation_flag** is used when $\alpha(h) > 1.5$. The **near_ideal_flag** is used when Skill(h) > 0 and $0.8 \leq \alpha(h) \leq 1.2$. The **low_sample_flag** indicates fewer than 30 verification samples for a model/horizon cell.

These thresholds are diagnostic conventions for this post-evaluation, not universal decision rules. A collapse flag does not invalidate a model, and an inflation flag does not imply a better dynamic representation; the flags are intended to make horizon-wise interpretation auditable and reproducible.

6. Reproducible Workflow

The five-model evaluation produces a unified station×model×horizon table with 595 rows (17 stations × 5 models × 7 horizons) containing persistence-relative RMSE skill, variance retention α , Skill_{VP}, and bootstrap confidence intervals (**alpha_ci_low**, **alpha_ci_high**). Repository scripts regenerate all manuscript tables and figures from the underlying forecast outputs, enabling inspection of post-evaluation assumptions (horizon structure, baseline definition, and preprocessing boundaries) and reproducible recomputation of the diagnostic layer [16].

Reproducibility here is intended to support the auditability of the evaluation workflow rather than introduce a new forecasting architecture: readers can re-run the study and examine how persistence-relative skill, horizon-wise behaviour, and variance retention interact.

7. Results

7.1. Five-model diagnostic summary

Across 17 stations and seven forecast horizons, direct HGB and Ridge achieve positive RMSE skill in all station-horizon cells but show near-universal variance collapse. HGB collapses in 118/119 cells (99.2%) and Ridge in 118/119 cells (99.2%), with only two $h = 1$ point-estimate exceptions: Huesca for HGB ($\alpha = 0.544$) and Barcelona Vall d’Hebron for Ridge ($\alpha = 0.502$). SARIMA exhibits a similar but slightly less extreme pattern, collapsing in 110/119 cells (92.4%). Seasonal naive preserves variance (0/119 collapsed cells) but has slightly negative median skill, whereas STL+Ridge avoids variance collapse but inflates variability and performs substantially worse in RMSE skill. Thus, the observed trade-off is not a simple model ranking problem: models can be significantly skillful in RMSE while dynamically over-smoothed, or dynamically variable while inaccurate.

Table 2: Model-family diagnostic summary across 17 stations and seven horizons. Collapse is defined as $\alpha < 0.5$. Near-ideal cells require positive RMSE skill and $\alpha \geq 0.8$. DM counts report BH-adjusted Diebold-Mariano tests against persistence. Exceedance metrics summarize event-detection behaviour across the evaluated thresholds.

Model	Family	Med. Skill	Med. α	Collapse %	Retained %	Near-Ideal %	Sig. Imp	Sig. Deg	Med. CSI	Med. FAR
hgb_direct	Direct ML	0.205	0.151	99.2%	0.0%	0.0%	111	0	0.121	0.647
ridge_direct	Direct ML	0.219	0.087	99.2%	0.0%	0.0%	117	0	0.104	0.595
sarima	Statistical baseline	0.208	0.095	92.4%	0.0%	0.0%	44	0	0.118	0.500
seasonal_naive	Variance-preserving naive	-0.026	1.000	0.0%	98.3%	16.8%	5	34	0.111	0.822
stl_ridge_direct	Decomposition + Ridge	-1.107	1.399	0.0%	100.0%	0.0%	0	119	0.104	0.896

Under the “near-ideal” criterion ($\text{Skill} > 0$ and $\alpha \geq 0.8$), no non-naive evaluated model attains any near-ideal cell: HGB direct, Ridge direct, SARIMA, and STL+Ridge each have 0/119 near-ideal cells. Seasonal naive contributes 20/119 near-ideal cells, but it is a naive reference and should not be interpreted as a superior predictive model.

7.2. Horizon profiles of skill and variance retention

To make the multi-horizon behaviour readable in the main text, we summarise horizon profiles using station-wise medians and interquartile ranges (IQR). Figure 1 reports median persistence-relative RMSE skill by horizon, and Figure 2 reports the corresponding variance-retention ratio α . The detailed three-model horizon table used in early diagnostic development is provided in the Supplementary Material.

7.3. Skill- α diagnostic space

Figure 3 provides a single-reading diagnostic summary of the skill-fidelity trade-off (cf. joint-performance summaries in [17]).

7.4. SARIMA and STL+Ridge as diagnostic controls

SARIMA and STL+Ridge were included to test whether the skill-variance trade-off is specific to direct lag-only ML. SARIMA behaves similarly to HGB and Ridge: it obtains positive median RMSE skill (0.208) but collapses in 110/119 cells (92.4%). STL+Ridge provides the opposite control case. It avoids variance collapse in all cells and has median $\alpha = 1.399$, but its median RMSE skill is strongly negative (-1.107) and all 119 station-horizon cells are significantly worse than persistence under the BH-adjusted DM test. This result shows that preserving amplitude is necessary but not sufficient: the retained variability must also be phase-aligned and forecast-relevant.

7.5. Sensitivity to the α threshold

The conclusion is robust to threshold perturbations. At $\alpha < 0.4$, HGB and Ridge still collapse in 92.4% and 89.9% of cells, respectively; at $\alpha < 0.5$ both collapse in 99.2%; and at $\alpha < 0.6$ both collapse in 100.0%. SARIMA also remains highly collapsed across this range (89.9%, 92.4%, and 96.6%, respectively). Seasonal naive and STL+Ridge remain non-collapsed at these thresholds. The threshold $\alpha = 0.5$ is therefore not driving the qualitative conclusion, although it should be interpreted as a diagnostic convention rather than a universal physical boundary.

7.6. Diebold–Mariano significance against persistence

The RMSE gains of the smoothed models are not merely sampling noise. Relative to persistence, the BH-adjusted Diebold–Mariano tests (paired squared losses with the Harvey–Leybourne–Newbold correction [18]) identify significant improvements in 111/119 HGB cells and 117/119 Ridge cells. SARIMA is significantly better in 44/119 cells. In contrast, STL+Ridge is significantly worse in all 119 cells. These tests make the central paradox explicit: statistically significant RMSE improvements can coexist with severe loss of predictive variance and poor event sensitivity.

7.7. Operational consequence: exceedance behaviour

The operational consequence of variance collapse is visible in event detection. For P75 events, mean recall is 26.9% for HGB, 21.0% for Ridge, and 15.6% for SARIMA, compared with 40.0% for persistence and 32.6% for seasonal naive. At P90, recall falls to 7.4%, 5.6%, and 6.9%, respectively. STL+Ridge achieves much higher recall (98.4% at P75 and 87.7% at P90), but with low precision (mean 24.2% at P75) and high false-alarm rates. Thus, neither RMSE skill nor variance retention alone is sufficient for operational forecasting: the evaluation must jointly consider amplitude, event detection, and false alarms.

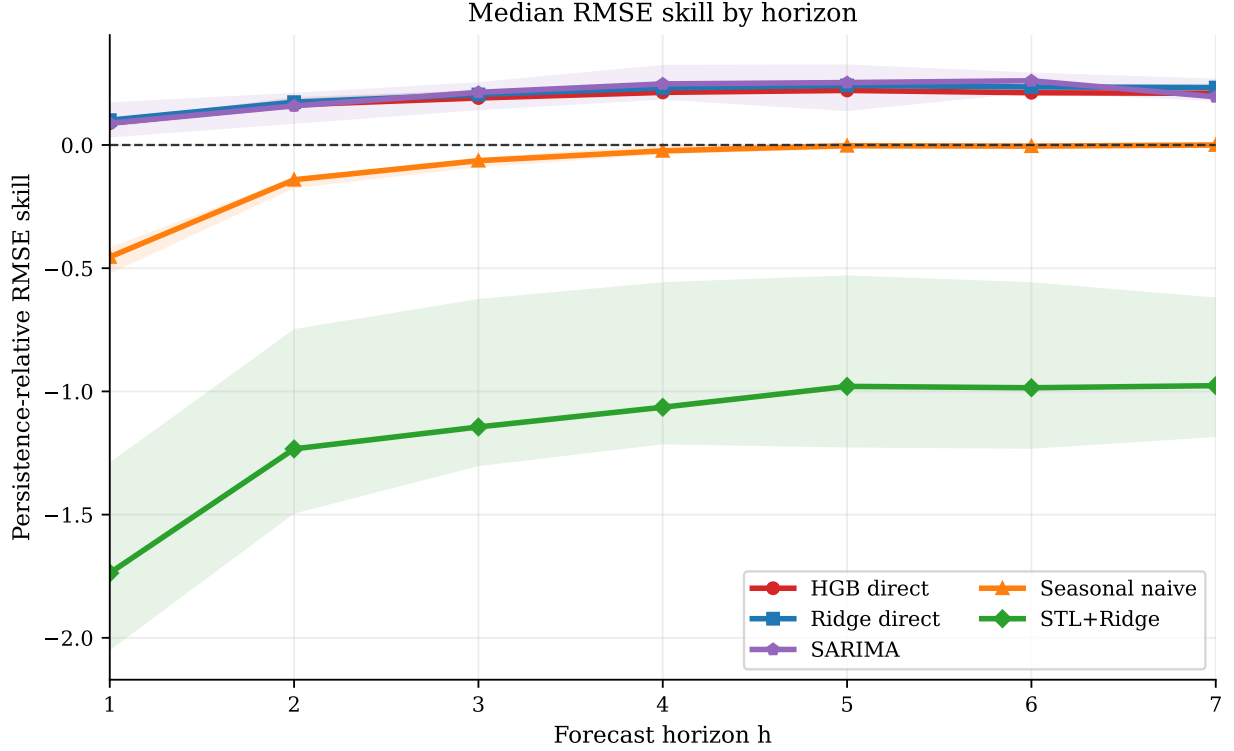


Figure 1: Median persistence-relative RMSE skill by forecast horizon across 17 PM_{10} stations. Shaded bands denote interquartile ranges across stations. HGB, Ridge, and SARIMA retain positive median skill, whereas seasonal naive remains close to persistence and STL+Ridge is substantially worse than persistence.

7.8. Murphy decomposition

The Murphy decomposition links the empirical findings to forecast verification theory [5]. The smoothed models reduce error while compressing predictive variability, whereas STL+Ridge preserves or inflates variability but introduces large error components. This decomposition supports the interpretation that the main failure mode is not simple unconditional bias but a mismatch between forecast amplitude, phase, and conditional structure.

7.9. Cross-station consistency and station heterogeneity

The station set is heterogeneous, including urban traffic, urban/background, suburban industrial, rural remote/background, and EMEP sites. Figure 7 shows the geographic distribution and confirms that variance collapse in direct ML is nearly ubiquitous across station types.

The results set an operational paradox: the direct ML forecasters can improve upon persistence in RMSE while producing trajectories that are substantially smoother than the observations. Interpreting performance therefore requires separating baseline-relative error reduction from the extent to which forecasts preserve day-to-day variability across horizons. The following discussion connects this decoupling to the squared-error objective and clarifies what it implies for episode-sensitive use.

8. Discussion

8.1. Positive skill does not imply dynamic fidelity

Daily PM_{10} forecasting exhibits a skill-fidelity trade-off: persistence-relative RMSE skill can be positive while point-forecast trajectories retain only a small fraction of observed day-to-day variability. In this

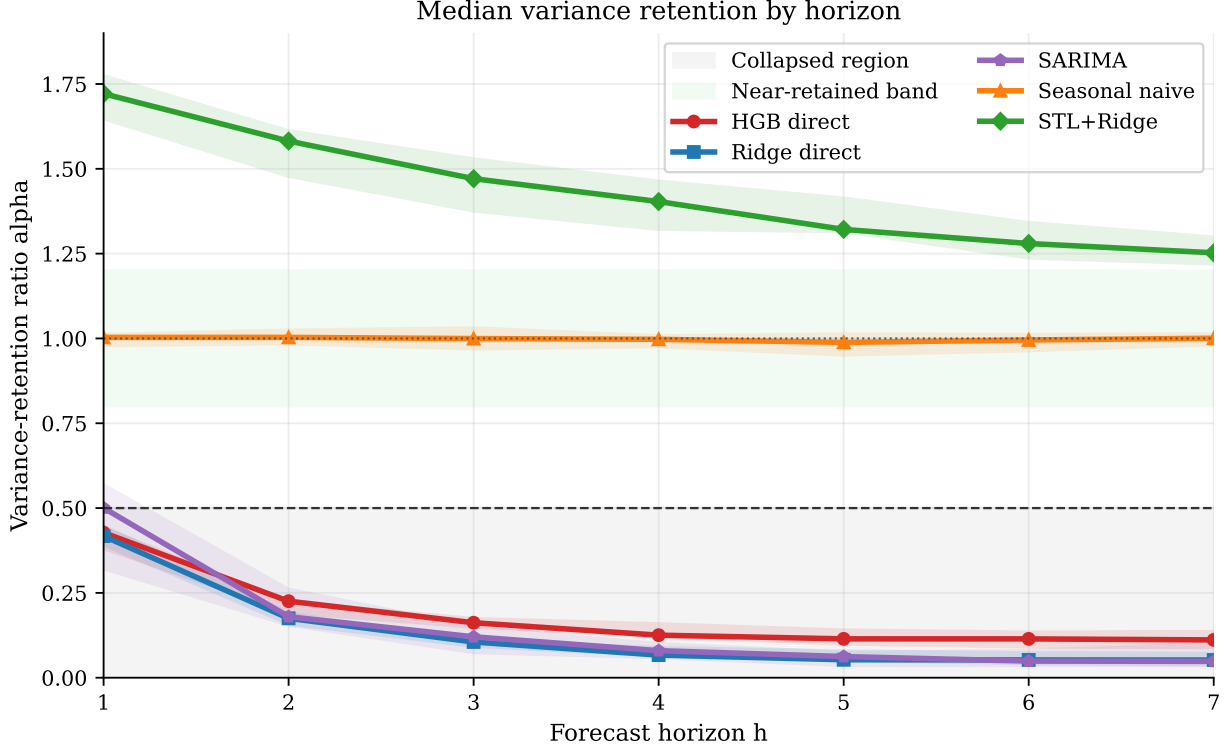


Figure 2: Median variance-retention ratio by forecast horizon across 17 PM_{10} stations. The grey region indicates collapsed variance ($\alpha < 0.5$), and the green band indicates near-retained variance. HGB, Ridge, and SARIMA rapidly enter the collapsed region, while seasonal naive remains close to $\alpha = 1$ and STL+Ridge preserves or inflates variance.

study, HGB direct and Ridge direct attain positive RMSE skill in all cells and are significantly better than persistence in most cells under the BH-adjusted DM test; yet, both collapse variance in 118/119 cells (99.2%) (Table 2).

8.2. Variance collapse is not restricted to machine learning

Including SARIMA shows that variance collapse is not an architectural defect unique to direct ML. SARIMA retains a positive median skill (0.208) but collapses in 110/119 cells (92.4%), indicating that the skill-fidelity tension can arise in classical statistical baselines as well as in machine-learning models when evaluated against persistence.

8.3. Variance preservation is not sufficient for usefulness

Seasonal naive preserves variance (0/119 collapsed cells; median $\alpha \approx 1.000$) but has a slightly negative median skill (-0.026) and a high median FAR (0.822). STL+Ridge preserves or inflates variance in all cells (median $\alpha = 1.399$) and shows very high exceedance recall, but its median RMSE skill is strongly negative (-1.107) and it exhibits high false-alarm rates (median FAR 0.896). Preserving amplitude is therefore necessary but not sufficient: variability must also be forecast-relevant and not dominated by false alarms.

8.4. Implications for operational PM_{10} forecasting

RMSE skill, variance retention, and exceedance behaviour expose different failure modes. Positive baseline-relative RMSE skill can coexist with variance collapse and trajectory smoothing, whereas variance preservation can coexist with weak RMSE skill and high false-alarm rates. Interpreting daily PM_{10} point forecasts therefore requires reading baseline-relative accuracy alongside variance retention and operational exceedance checks, rather than treating any single summary as a universal score.

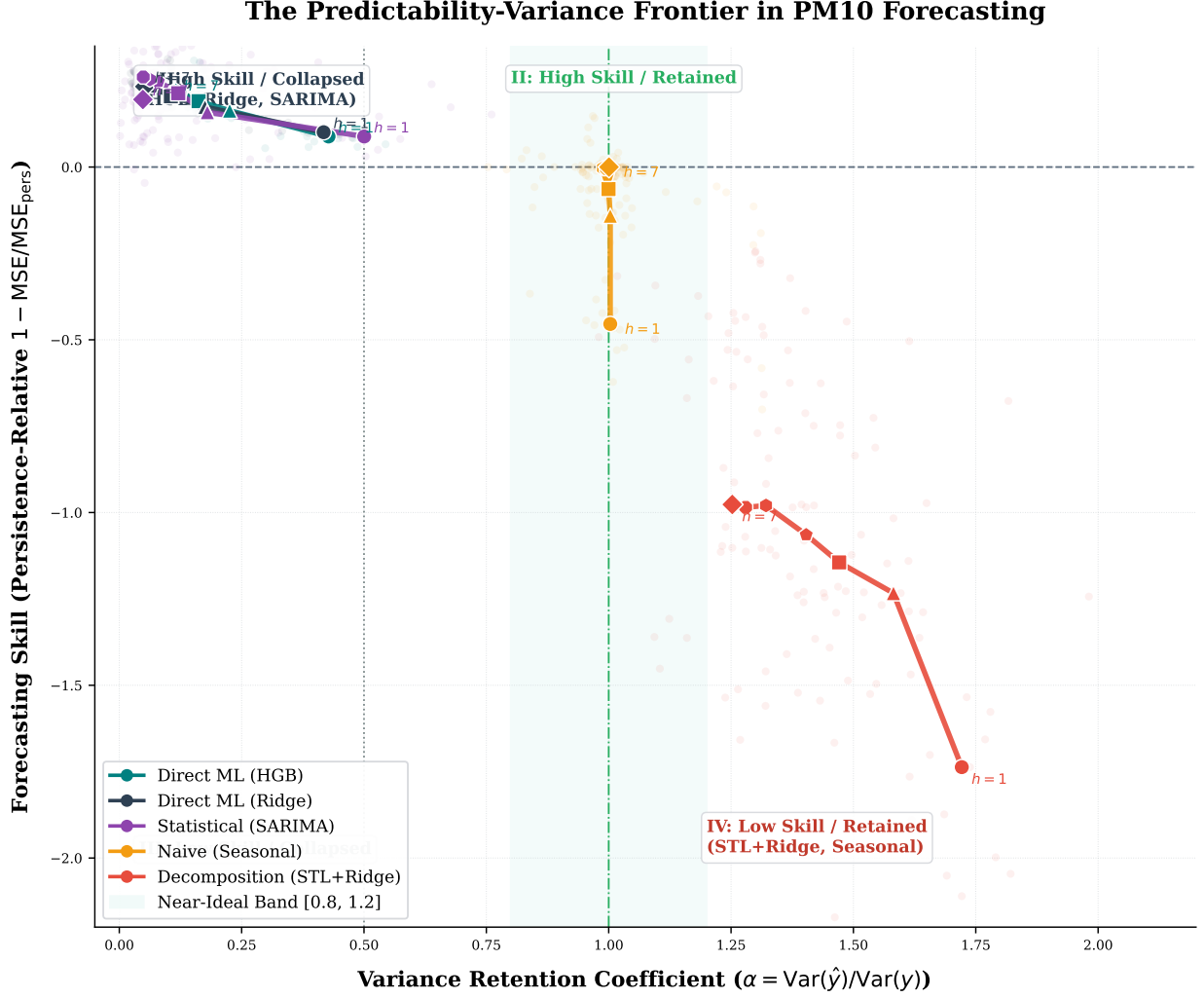


Figure 3: Skill-retention diagnostic by model. Points show model-level medians across station-horizon cells, and error bars show interquartile ranges. HGB, Ridge, and SARIMA occupy the skillful-but-collapsed region, seasonal naive lies near retained variance but near-zero skill, and STL+Ridge preserves or inflates variance while losing RMSE skill.

9. Implications for Forecast Evaluation Practice

Minimum reporting practices can substantially improve interpretability in PM forecasting evaluation: (i) report persistence-relative RMSE skill so that gains are anchored to an operational baseline; (ii) report horizon-wise performance rather than only aggregate scores; (iii) disclose temporal validation design and train-only preprocessing boundaries to support deployment-like credibility; and (iv) when point trajectories are interpreted operationally, report variance retention or another trajectory-fidelity diagnostic as a complementary post-evaluation output.

10. Limitations

This study is intentionally limited in scope. The empirical evidence covers 17 MITECO stations at the daily temporal scale; transfer to other pollutants, temporal resolutions, or national networks is not assessed. The five-model comparison is diagnostic rather than exhaustive; the models are intended to reveal

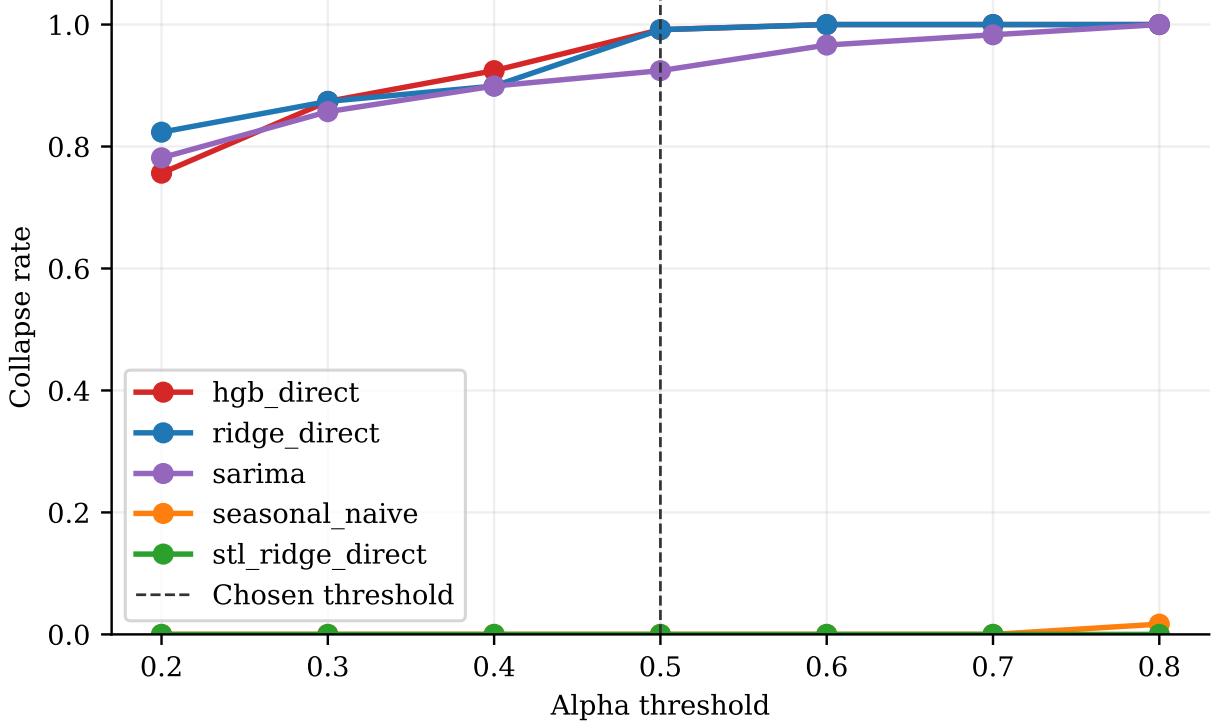


Figure 4: Sensitivity of the variance-collapse diagnostic to the α threshold. Collapse remains near-universal for HGB and Ridge across reasonable thresholds around $\alpha = 0.5$, and SARIMA follows the same qualitative pattern. Seasonal naive and STL+Ridge remain non-collapsed over the same range, showing that the diagnostic separates smoothed forecasts from variance-preserving or variance-inflating alternatives.

failure modes under a common rolling-origin protocol, not to establish a definitive leaderboard. The model set spans two direct lag-feature ML forecasters (HGB direct, Ridge direct), a sparse classical statistical baseline (SARIMA), a decomposition-based variant (STL+Ridge), and a seasonal naive reference; other architectures, covariate-rich designs, and probabilistic forecasters may exhibit different variance-retention behaviour. SARIMA is used as a sparse statistical baseline; its results should not be interpreted as the best attainable SARIMA performance under extensive manual specification. Exceedance diagnostics are included as compact operational checks, not as a full event-classification benchmark. Skill_{VP} is not proposed as a general-purpose replacement for RMSE or as a new standard verification score; it is a simple diagnostic auxiliary whose value lies in making the skill-fidelity decoupling explicit at each horizon. The collapse threshold $\alpha = 0.5$ is operationally motivated but not derived from a formal loss criterion. It should be interpreted as a diagnostic convention rather than a universal physical boundary. Threshold-sensitivity results for the five-model experiment (Figure 4) support the same qualitative conclusion across reasonable perturbations around $\alpha = 0.5$.

11. Conclusion

Daily PM₁₀ forecasting exhibits a skill-fidelity trade-off under persistence-relative RMSE evaluation. HGB direct, Ridge direct, and SARIMA achieve positive median persistence-relative RMSE skill, but they do so while collapsing forecast variance (median α 0.151, 0.087, and 0.095; collapse rates 99.2%, 99.2%, and 92.4%, respectively). Seasonal naive and STL+Ridge preserve or inflate variance (median $\alpha \approx 1.000$ and 1.399) but do not provide useful RMSE skill on median (median skill -0.026 and -1.107), and STL+Ridge

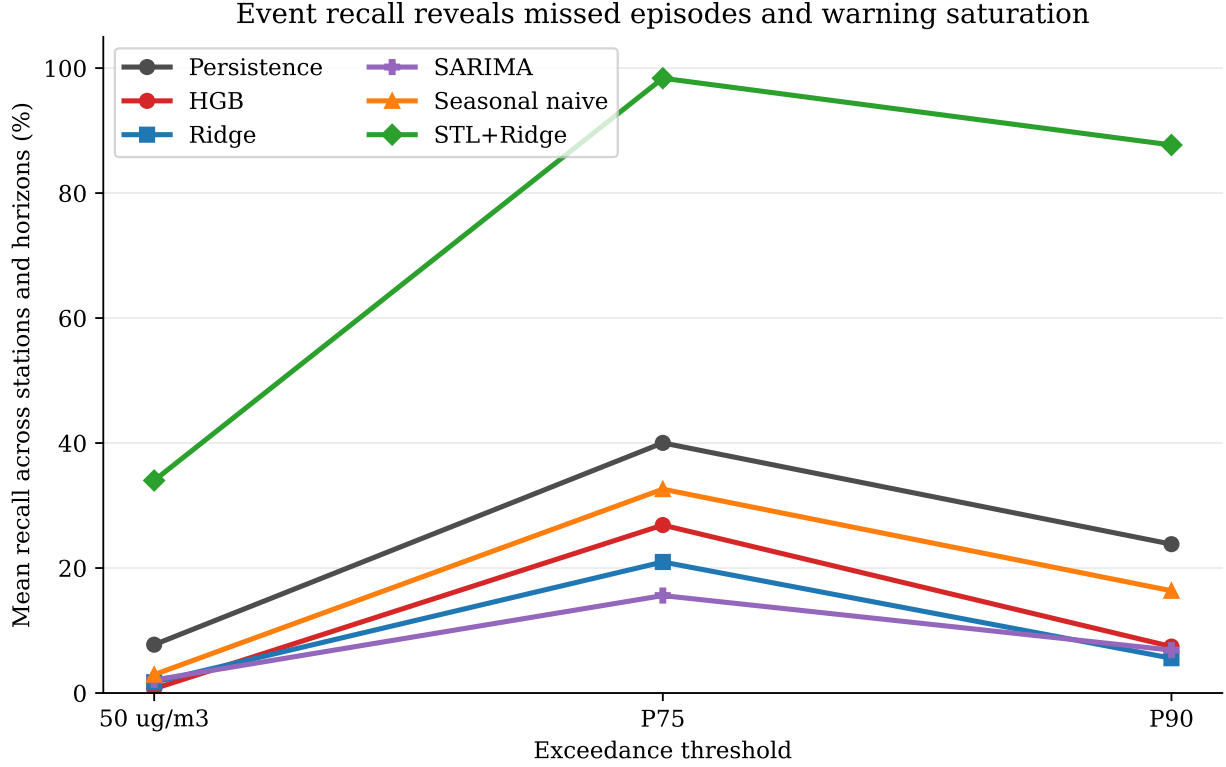


Figure 5: Mean exceedance recall across stations and horizons. Despite positive RMSE skill, HGB, Ridge, and SARIMA miss many P75 and P90 events. STL+Ridge attains high recall, but this must be interpreted together with its low precision and high false-alarm rate.

illustrates that variance preservation can coincide with low precision and high false-alarm rates in exceedance diagnostics (Table 2).

The central lesson is therefore not that one model family is universally preferable, but that baseline-relative accuracy, variance retention, and exceedance behaviour expose different failure modes. In particular, statistically significant RMSE gains (as confirmed by BH-adjusted Diebold–Mariano tests) can coexist with severe variance collapse and weak event sensitivity, while variance preservation can coexist with poor RMSE skill and high false-alarm rates. Skill_{VP} is retained only as an auxiliary diagnostic to compactly summarise the joint reading of skill and variance retention, not as a universal verification score. A forecast that beats persistence in RMSE is not necessarily dynamically credible, and a forecast that preserves variance is not necessarily operationally useful.

CRediT author statement

Federico García Crespi: Conceptualisation, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualisation, Writing – original draft, Writing – review & editing. **Julio Alberto Ramos Martínez:** Conceptualisation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

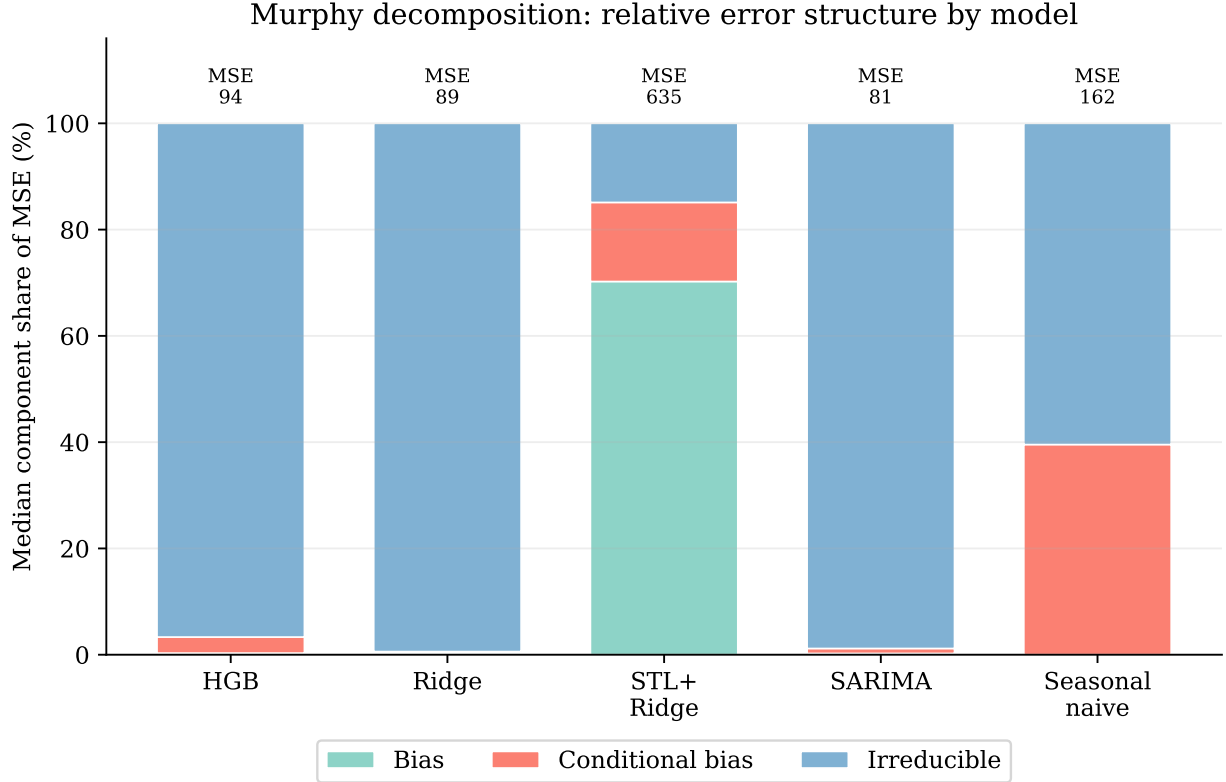


Figure 6: Murphy decomposition of mean squared error by model. Bars show median percentage contribution of bias, conditional bias, and irreducible components; labels above bars report median MSE. The decomposition shows that variance preservation alone does not guarantee useful deterministic forecasts.

Data Availability Statement

Raw daily PM_{10} observations were obtained from the MITECO Spanish national air-quality network portal (<https://www.miteco.gob.es/>) and are publicly available. The analysis code, processed per-station series, and the 595-cell aggregate diagnostic tables (variance retention, Diebold–Mariano significance, exceedance, Murphy decomposition) underlying every multi-station figure are version-controlled at <https://github.com/fedeg-umh-es/varret-pm10-paper> under `evidence/paper_a/`, with SHA-256 checksums and provenance recorded in `evidence/paper_a/manifests/provenance_manifest.csv`. The row-level prediction set (895,737 rows spanning all 17 stations, 5 model families, and 7 horizons) is available as a versioned GitHub release asset of the same repository (tag `paper-a-row-level-evidence-v1`), with its SHA-256 checksum reported in `evidence/paper_a/row_level/predictions_all_stations_manifest.md` and in the release notes. This row-level file directly supports independent recomputation of variance retention (α), collapse, and near-ideal classification for all 595 station-model-horizon cells, and of persistence-relative RMSE skill for the four non-SARIMA model families; SARIMA’s cell-level skill was computed against a wider persistence baseline (`skill_sarima_{station}.csv`) not included in this row-level release, as documented in the manifest. The Supplementary Material contains the PRISMA reporting-audit outputs and the full monitoring-station table referenced in the main text.

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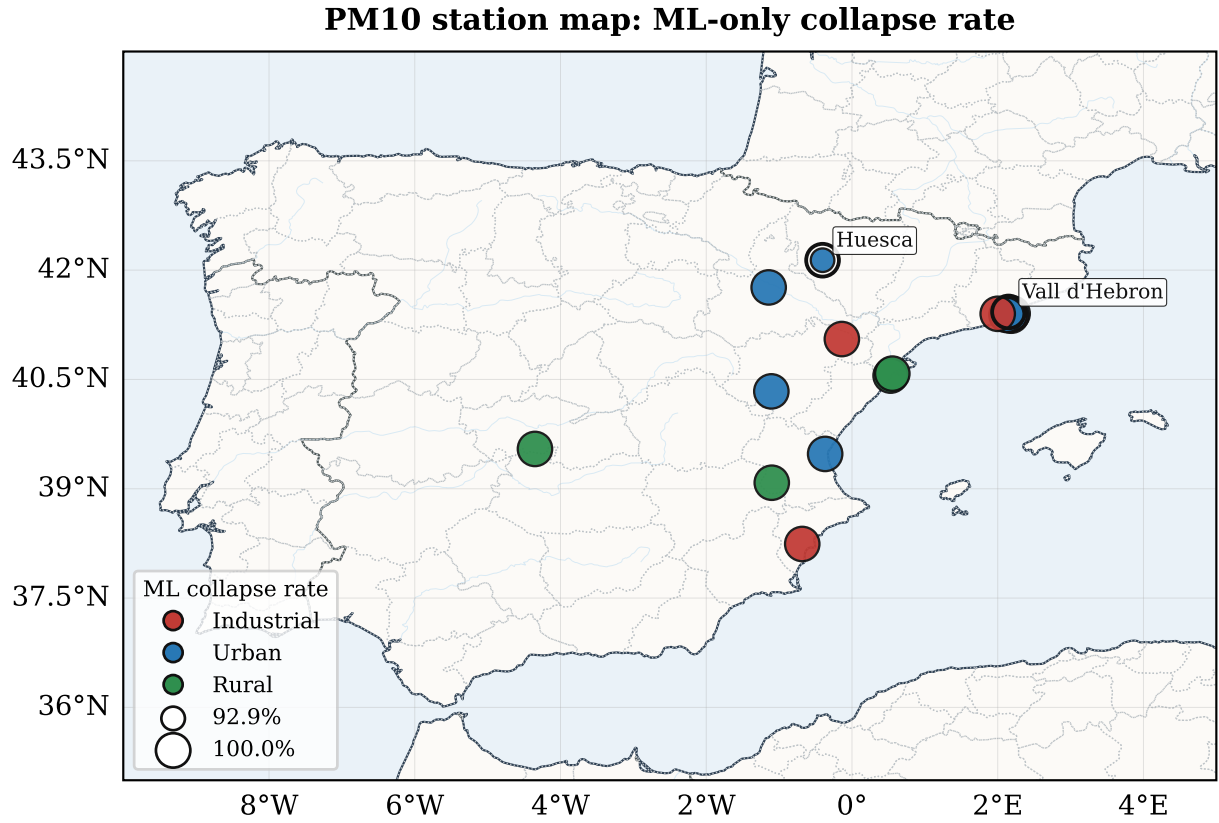


Figure 7: Geographic distribution of the 17 MITECO stations. Marker colour denotes station type. Marker size denotes the ML-only collapse rate, computed over 14 model-horizon cells per station (2 direct ML models x 7 horizons). Fifteen stations show 14/14 collapsed ML cells; Huesca and Barcelona Vall d'Hebron show 13/14 due to one non-collapsed h=1 cell each.

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Supplementary Material

S1. PRISMA reporting audit (supporting motivation)

Table 3: Abstract-level reporting visibility of evaluation-relevant practices in PM forecasting studies. Percentages are computed over 486 abstract-coded studies; low reporting rates indicate reporting opacity, not confirmed absence of the corresponding practice in full text.

Evaluation dimension	Explicitly reported	%	Interpretive function
Persistence or naive baseline	18/486	3.7	Minimum operational comparator
Horizon-wise reporting	11/486	2.3	Multi-step interpretability
Temporally appropriate validation	24/486	4.9	Deployment-like temporal credibility
Explicit leakage-safe safeguards	3/486	0.6	Protection against information bleed
Operational-usefulness framing	51/486	10.5	Decision relevance

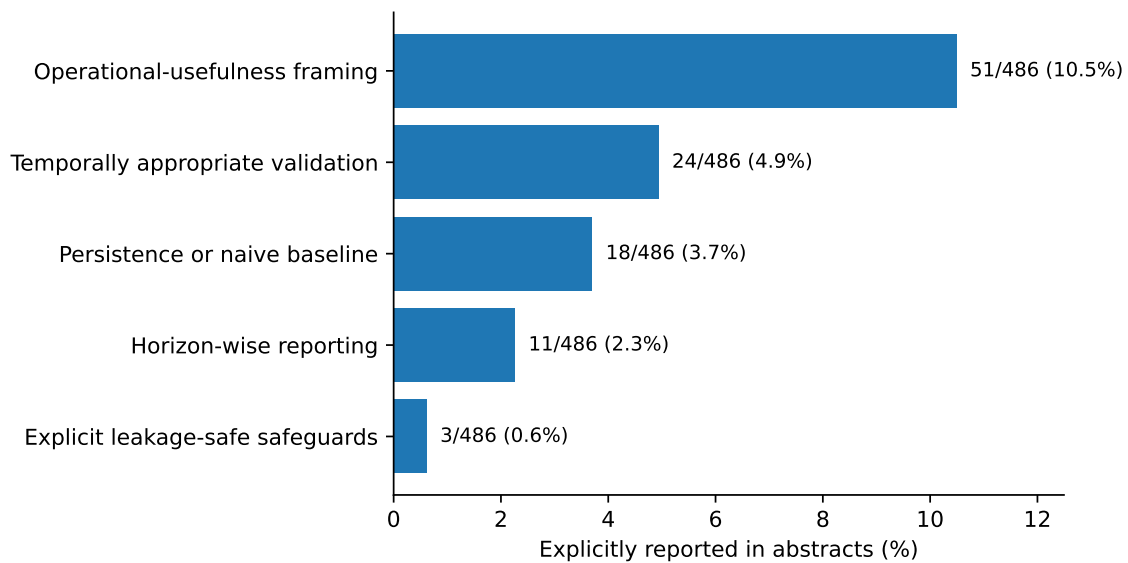


Figure 8: Abstract-level reporting visibility of key evaluation practices in particulate matter forecasting studies. Percentages are computed over 486 abstract-coded studies. Low percentages indicate reporting opacity, not confirmed absence of the corresponding practice in the full text.

S2. Monitoring stations

Table 4: Monitoring stations included in the study. Stations span industrial, traffic, urban background, suburban, and rural remote settings within the MITECO national air-quality network. Coverage refers to available daily PM₁₀ observations within the study period (2017–2024).

Station code	Name	Area type	Altitude (m a.s.l.)	Coverage (%)
ES0446A	Elx-Agroalimentari	Suburban/Industrial	44	80.4
ES1619A	Valencia-Vivers	Urban/Residential	11	91.7
ES0012R	Zarra-EMEP	Rural Remote EMEP	855	96.0
ES1870A	Barcelona El Port Vell	Suburban/Industrial	1	98.5
ES1418A	Alagon	Suburban/Traffic	235	97.5
ES1421A	Teruel	Urban/Background	915	96.7
ES0001R	San Pablo de los Montes	Rural	923	96.1
		Remote/Background		
ES1438A	Barcelona L'Eixample	Urban/Traffic	26	95.3
ES0567A	Barcelona Zona Universitaria	Urban/Background	24	95.2
ES1417A	Huesca	Urban/Traffic	488	95.0
ES0691A	Barcelona El Poblenou	Urban/Background	3	94.9
ES0559A	Barcelona Pl. Universitat	Urban/Traffic	24	94.9
ES1856A	Barcelona Vall d'Hebron	Urban/Background	136	94.5
ES1966A	Alcaniz Capuchinos	Urban/Industrial	340	94.3
ES2011A	Sant Vicenc dels Horts	Suburban/Industrial	65	94.1
	Alaba			
ES2091A	Alcanar Montecarlo	Rural	6	93.7
		Remote/Industrial		
ES2017A	Alcanar Llar de Jubilats	Rural	7	93.6
		Near-city/Industrial		

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